

# Experiment 10: Strategy Evaluation – Perform a Comparative Analysis of LLM Responses Across Diverse Prompting Methodologies

## Aim

To compare the responses of a Large Language Model (LLM) using different prompting methodologies such as Zero-shot, One-shot, Few-shot, Chain-of-Thought, and Meta Prompting.

## Python Code:

```
from openai import OpenAI
```

```
client = OpenAI(
    api_key="sk-proj-Sd2jzBsSZTqHY1pDWZLea6fUdkzFa-
vQyeUTL2Kdmb1oO8AFZReX80XQrD-
eDbfpOUyzzDnvnDT3BlbkFJP3FNYcAR4S6TPPtKO-Uw6osECufNvkaZ3kXB-
WF1xw2XCykfyiZLxwUzLGpzfESUsNJGxcc6YA"
)
```

```
prompts = {
    "Zero-shot":
        "Explain the importance of Artificial Intelligence in education.",
```

```
    "One-shot":
        """Example:
        Artificial Intelligence in healthcare helps doctors diagnose diseases accurately.
```

```
        Now explain the importance of Artificial Intelligence in education.""",
```

```
    "Few-shot":
        """Example 1:
        AI in healthcare improves diagnosis and patient care.
```

```
        Example 2:
        AI in banking detects fraud and improves security.
```

```
        Now explain the importance of Artificial Intelligence in education.""",
```

```
    "Chain-of-Thought":
        """Think step by step and explain the importance of Artificial Intelligence in
        education.""",
```

```
    "Meta Prompting":
        """First create the best prompt to explain the importance of Artificial Intelligence in
        education. Then answer using that improved prompt."""
```

```

}

for name, prompt in prompts.items():
    print(f"\n===== {name} =====\n")

    response = client.responses.create(
        model="gpt-5.4-mini",
        input=prompt
    )

    print(response.output_text)

```

## Output

===== **Zero-shot** =====

Artificial Intelligence (AI) is important in education because it can make learning more personalized, efficient, and accessible for students and teachers.

### ### 1. Personalized learning

AI can adapt lessons to each student's level, pace, and learning style. For example, if a student is struggling with math, an AI-based system can provide extra practice and simpler explanations. If another student is advanced, it can offer more challenging material.

### ### 2. Immediate feedback

AI tools can quickly check answers, quizzes, and assignments, giving students instant feedback. This helps them understand mistakes right away and improve faster.

### ### 3. Support for teachers

AI can automate time-consuming tasks like grading, attendance, and organizing materials. This gives teachers more time to focus on teaching, mentoring, and supporting students.

### ### 4. Better accessibility

AI can help students with disabilities by offering features like speech-to-text, text-to-speech, translation, and reading assistance. This makes education more inclusive.

### ### 5. Data-driven insights

AI can analyze student performance and identify patterns, such as who may be falling behind or which topics are difficult for many learners. Teachers and schools can use this information to improve instruction.

### ### 6. Learning anytime, anywhere

AI-powered chatbots, tutoring systems, and learning apps can help students study outside the classroom, making education more flexible and available.

### ### 7. Preparation for the future

Since AI is becoming part of many industries, using it in education helps students develop digital and problem-solving skills that are valuable for future careers.

### ### Conclusion

AI is important in education because it improves learning outcomes, supports teachers, and makes education more inclusive and efficient. When used responsibly, it can greatly enhance the teaching and learning experience.

## ===== **One-shot** =====

Artificial Intelligence in education helps teachers personalize learning, identify student difficulties early, and make learning more effective and accessible.

It can:

- adapt lessons to each student's pace and level,
- provide instant feedback on assignments,
- automate routine tasks for teachers,
- support students with special learning needs,
- and make education available anytime through smart learning tools.

Overall, AI improves teaching, saves time, and helps students learn better.

## ===== Few-shot =====

Artificial Intelligence in education helps personalize learning, support teachers, and improve student performance.

It can adapt lessons to each student's pace, provide instant feedback, automate grading, and identify learning gaps early. AI also makes education more accessible through tools like tutoring systems, translation, and speech recognition.

## ===== Chain-of-Thought =====

Artificial Intelligence (AI) is becoming very important in education because it helps make learning more effective, personalized, and accessible. Here's a step-by-step explanation of why it matters:

### 1. \*\*Personalized learning\*\*

- AI can adapt lessons to each student's pace and skill level.
- This means strong students can move ahead faster, while others can get extra help where they struggle.

### 2. \*\*24/7 support\*\*

- AI-powered chatbots and virtual tutors can answer questions anytime.
- Students do not have to wait for class time or office hours to get help.

### 3. \*\*Better feedback\*\*

- AI can quickly check quizzes, assignments, and practice exercises.
- Students receive immediate feedback, which helps them improve faster.

### 4. \*\*Helps teachers save time\*\*

- Teachers spend a lot of time on grading and administrative tasks.
- AI can automate some of these tasks, giving teachers more time to focus on teaching and supporting students.

#### 5. **Supports students with special needs**

- AI can provide tools like speech-to-text, text-to-speech, translation, and reading assistance.
- This makes education more inclusive for learners with disabilities or language barriers.

#### 6. **Improves access to education**

- AI-based platforms can deliver learning resources online to students in remote areas.
- This helps more people get quality education, even when schools or teachers are limited.

#### 7. **Data-driven insights**

- AI can analyze learning patterns and identify where students are having difficulty.
- Teachers and schools can use this information to improve lessons and support students earlier.

### ### Why this is important overall

AI helps education become:

- more **personalized**
- more **efficient**
- more **inclusive**
- more **accessible**

### ### Conclusion

Artificial Intelligence is important in education because it improves the learning experience for students and makes teaching easier for educators. When used wisely, AI can help create a smarter, fairer, and more effective education system.

If you want, I can also turn this into:

- a **short paragraph**
- a **5-mark answer**
- or a **speech/presentation format**

## ===== Meta Prompting =====

**\*\*Improved prompt:\*\***

“Explain the importance of Artificial Intelligence in education in a clear, balanced, and engaging way. Cover how AI supports personalized learning, saves teachers time, improves accessibility, and helps schools make better decisions. Also mention possible challenges, such as privacy concerns and overreliance on technology, and end with a brief conclusion about AI’s future role in education.”

**\*\*Answer:\*\***

Artificial Intelligence is becoming increasingly important in education because it can make learning more personalized, efficient, and accessible. One of its biggest advantages is personalized learning. AI systems can analyze a student’s progress and adapt lessons to fit their strengths, weaknesses, and learning pace. This helps students get the support they need instead of following a one-size-fits-all approach.

AI also helps teachers by automating time-consuming tasks such as grading, tracking student performance, and organizing learning materials. As a result, teachers can spend more time focusing on teaching, mentoring, and supporting students individually.

Another major benefit is accessibility. AI tools can assist students with disabilities through features like speech-to-text, text-to-speech, translation, and adaptive interfaces. This makes education more inclusive and available to a wider range of learners.

In addition, AI can help schools and educators make better decisions by analyzing large amounts of data. It can identify learning trends, predict where students may struggle, and support early intervention.

However, there are also challenges. Schools must protect student privacy and ensure that AI is used responsibly. There is also a risk that students and teachers may rely too much on technology, which could reduce human interaction in learning.

Overall, AI has the potential to transform education by making it more effective, inclusive, and flexible. When used wisely, it can support both teachers and students and play a major role in the future of learning.

## **Result**

The responses generated using different prompting methodologies were successfully compared. Meta Prompting and Chain-of-Thought produced the most detailed and accurate responses, while Few-shot prompting also achieved high-quality results. Zero-shot and One-shot prompting generated acceptable responses with comparatively less detail.